

ShopSense

E-Commerce Customer Segmentation Using Machine Learning

Project Report
Unsupervised customer segmentation using K-Means and Agglomerative Hierarchical Clustering

This report is based on the uploaded customer dataset, problem statement, and the exported analysis notebook. All numerical results and cluster interpretations are taken from the notebook output.

Project Element	Notebook Result
Original dataset	2,240 customers × 22 attributes
Missing values	24 values in Income
Post-outlier dataset	2,236 customers
Encoded clustering matrix	2,236 × 18 features
PCA representation used for clustering	3 principal components
Selected number of clusters	4
Algorithms	K-Means + Agglomerative (Ward linkage)
Final cluster characterisation	Agglomerative cluster labels

Source note: The uploaded notebook contains the complete preprocessing, PCA, K selection, clustering, visualisation and cluster-summary workflow.

1. Project Overview

ShopSense is an e-commerce customer segmentation project designed to identify meaningful groups of customers from historical customer data. The source problem statement describes 2,240 customer records and 22 attributes covering demographics, purchase behaviour, website activity and response. The business objective is to group customers according to similarities in purchasing behaviour, engagement levels and loyalty indicators so that the resulting segments can support data-driven marketing and customer-retention decisions.

2. Problem Statement

The source problem statement notes that generic marketing and engagement strategies are applied across customers without clearly understanding different customer behaviour patterns. The stated consequences include inefficient marketing, missed opportunities to retain high-value customers, and delayed identification of churn-prone users. ShopSense addresses this by applying unsupervised clustering to discover hidden customer-behaviour patterns.

3. Dataset

Feature Group	Variables
Customer demographics	ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome, Dt_Customer
Purchase amount	MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds
Purchase frequency	NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth
Customer feedback / response	Recency, Complain, Response

4. Data Preprocessing & Feature Engineering

Missing values: The notebook identifies 24 missing values in Income. These were replaced using the median income.

Engineered features: Age was calculated as 2026 minus Year_Birth; Customer_Tenure_Days was calculated from Dt_Customer using the maximum Dt_Customer in the dataset as the reference date; Total_Spending was created by summing the six product-spending variables; and Total_Children was created by adding Kidhome and Teenhome.

Categorical transformation: Education was consolidated into Graduate, Postgraduate and Undergraduate. Marital_Status was transformed into Living_With with Partner and Single categories. One-hot encoding was then applied to Education and Living_With.

Feature removal: The notebook dropped ID, Year_Birth, Marital_Status, Kidhome, Teenhome, Dt_Customer and the six original product-spending columns after creating the corresponding engineered/derived representation.

5. Outlier Treatment

Outliers were examined using Income, Recency, Response, Age, Total_Spending and Total_Children. The notebook retained observations satisfying Age < 90 and Income < 600,000.

Stage	Number of Records
Before outlier removal	2,240
After outlier removal	2,236
Observations removed	4

6. Encoding and Scaling

After one-hot encoding the Education and Living_With variables, the clustering feature matrix contained 18 features. StandardScaler was then applied so that features were placed on a common scale before the distance-based clustering analysis.

7. Principal Component Analysis (PCA)

PCA was used to reduce the scaled feature space and provide a compact representation for visualisation and clustering.

PCA Representation	Explained Variance
PCA 1	23.163%
PCA 2	11.385%
PCA 3	10.406%
First 2 components combined	34.548%
First 3 components combined	45.154%

The notebook subsequently used the three-component PCA representation (PCA-1, PCA-2 and PCA-3) as the input for the clustering models.

8. Selecting the Number of Clusters

The notebook evaluated K values from 1 to 10 using the Elbow Method and calculated WCSS (within-cluster sum of squares) for each K. KneeLocator identified the elbow at K = 4. Silhouette scores were additionally calculated for K = 2 through K = 10 as a supplementary clustering diagnostic.

Selection statement: The notebook explicitly reports Optimal K: 4 from the KneeLocator/Elbow analysis. The silhouette analysis is presented as an additional diagnostic rather than as the criterion that selected K=4.

9. Clustering Algorithms

K-Means: KMeans was fitted with n_clusters=4 and random_state=42 on the three-component PCA representation.

Agglomerative Hierarchical Clustering: AgglomerativeClustering was fitted with n_clusters=4 and linkage='ward' on the same PCA representation. The notebook then used the Agglomerative labels for the final cluster characterisation.

10. Final Cluster Characterisation

The final cluster summary is the mean of the variables in the analysis matrix grouped by the Agglomerative cluster labels. The cluster IDs 0-3 are algorithm-generated identifiers and are not rankings.

Metric	C0	C1	C2	C3
Income	39,680.58	72,808.45	36,960.14	70,722.68
Recency	48.91	49.20	48.32	50.50
Deal Purchases	2.59	1.96	2.59	1.86
Web Purchases	3.15	5.69	2.71	5.79
Catalogue Purchases	0.97	5.50	0.84	5.01
Store Purchases	4.14	8.66	3.62	8.43
Web Visits / Month	6.31	3.58	6.66	3.73
Complain	0.0111	0.0056	0.0113	0.0057
Response	7.62%	16.67%	14.19%	32.01%
Age	55.67	59.49	55.69	58.93
Customer Tenure (days)	342.94	369.72	338.78	376.28
Total Spending	221.96	1,236.59	165.70	1,190.39
Total Children	1.24	0.51	1.27	0.46
Living With Partner	100.00%	100.00%	0.68%	0.00%
Living With Single	0.00%	0.00%	99.32%	100.00%

11. C0 — Family-Oriented Lower-Spend Customers

C0 has average income of 39,680.58 and total spending of 221.96. It records 6.31 website visits per month, 3.15 web purchases, 4.14 store purchases and a 7.62% campaign response rate. Average total children is 1.24. The profile therefore combines relatively higher household-child counts and website activity with comparatively lower spending and campaign response.

Business implication: Focus on improving conversion and campaign engagement through family-oriented offers, personalised recommendations and retargeting.

12. C1 — High-Spending Multi-Channel Customers

C1 has the highest average income (72,808.45) and the highest average total spending (1,236.59). It also records strong purchasing frequency across web (5.69), catalogue (5.50) and store (8.66) channels. The average campaign response is 16.67%, while average total children is 0.51.

Business implication: Loyalty, repeat-purchase and cross-selling initiatives are aligned with this group's established multi-channel purchasing behaviour.

13. C2 — Digital-Engaged Lower-Spend Customers

C2 has the lowest average income (36,960.14) and the lowest total spending (165.70), while recording the highest website-visit frequency at 6.66 visits per month. Web purchases average 2.71 and store purchases 3.62. The Living_With variable shows 99.32% classified as Single. Campaign response averages 14.19%.

Business implication: Digital-first engagement and conversion initiatives are appropriate for this segment, particularly personalised online offers and retargeting.

14. C3 — High-Value Highly Responsive Customers

C3 has high average income (70,722.68) and total spending (1,190.39), together with strong purchasing activity across web (5.79), catalogue (5.01) and store (8.43) channels. It has the highest campaign response rate at 32.01%, the lowest average total children at 0.46, and the longest average customer tenure at 376.28 days. All customers in this cluster are classified as Single in the Living_With transformation.

Business implication: Retention, premium service, personalised engagement and high-value cross-selling can be considered for this highly responsive, high-spending group.

15. Comparative Customer Insights

Observation	Evidence from Cluster Means
Higher-value purchasing groups	C1 = 1,236.59 and C3 = 1,190.39 total spending; C0/C2 are 221.96 and 165.70.
Highest digital browsing	C2 has the highest Web Visits / Month at 6.66, followed by C0 at 6.31.
Strongest campaign response	C3 = 32.01%, followed by C1 = 16.67%, C2 = 14.19%, C0 = 7.62%.
Store-oriented high purchasing	C1 = 8.66 and C3 = 8.43 average store purchases.
Household composition	C0 = 1.24 and C2 = 1.27 average total children; C1 = 0.51 and C3 = 0.46.
Digital engagement vs. spending	C2 combines the highest website-visit frequency with the lowest average total spending.

16. Segment-Specific Business Actions

Segment	Primary Focus	Suggested Actions
C0	Improve conversion	Family bundles, personalised recommendations, retargeting, cart recovery
C1	Loyalty and repeat value	Loyalty rewards, repeat-purchase incentives, cross-selling, omnichannel engagement
C2	Digital conversion	Digital coupons, personalised web offers, retargeting, cart-abandonment campaigns
C3	Retention and value	Premium loyalty, exclusive offers, personalised service, high-value cross-selling

17. Visual and Analytical Outputs in the Notebook

The notebook includes pair plots for selected variables, a correlation heatmap, 2D and 3D PCA visualisations, the Elbow/WCSS curve, silhouette-score analysis, a combined WCSS and silhouette plot, 3D K-Means and Agglomerative cluster visualisations, a cluster-count plot, an Income-versus-Total-Spending scatter plot, and a final cluster-summary table.

18. Conclusion

The ShopSense analysis applies a complete unsupervised-learning workflow to e-commerce customer data: missing-value treatment, feature engineering, categorical consolidation and encoding, outlier treatment, standardisation, PCA, cluster-count analysis, and two clustering algorithms. The Elbow/Knee analysis selects four clusters, and both K-Means and Agglomerative Hierarchical Clustering are implemented with four clusters. The final characterisation uses the Agglomerative labels and reveals four distinct customer profiles that differ particularly in income, total spending, purchasing channels, website activity, household composition and campaign response.

These findings provide a data-driven basis for segment-specific customer engagement. In particular, C2 shows high website activity but low spending, C1 and C3 show substantially higher spending and multi-channel purchasing, and C3 has the strongest campaign response. These differences can be used to tailor marketing, loyalty, digital conversion and retention strategies while keeping the cluster IDs as descriptive identifiers rather than rankings.

Source: Uploaded ShopSense/SmartCart clustering notebook and problem-statement documents. The report intentionally excludes the separate income-spending segmentation as requested and focuses on the ML clustering analysis.